

# Node-Pruning Comparison

Normal node pruning at two operating points versus the final position-aware checkpoint

## Circuit sizes

| Granularity / size measure  | KL-match open/total    | KL closed | Epoch-match open/total | Epoch closed | Position open/total    | Position closed |
|-----------------------------|------------------------|-----------|------------------------|--------------|------------------------|-----------------|
| High-level abstract edges   | 2,008/11,754           | 82.9%     | 1,703/11,754           | 85.5%        | 1,276/109,722          | 98.8%           |
| Attention blocks            | 11/12                  | 8.3%      | 10/12                  | 16.7%        | 13/108                 | 88.0%           |
| Attention heads             | 51/144                 | 64.6%     | 48/144                 | 66.7%        | 75/1,296               | 94.2%           |
| Attention head neurons      | 2,631/9,216            | 71.5%     | 2,412/9,216            | 73.8%        | 3,818/82,944           | 95.4%           |
| MLP blocks                  | 12/12                  | 0.0%      | 10/12                  | 16.7%        | 25/108                 | 76.9%           |
| MLP hidden neurons          | 6,082/36,864           | 83.5%     | 4,319/36,864           | 88.3%        | 14,701/331,776         | 95.6%           |
| MLP output neurons          | 3,119/9,216            | 66.2%     | 2,344/9,216            | 74.6%        | 7,265/82,944           | 91.2%           |
| Whole-model parameter proxy | 54,532,773/124,439,808 | 56.2%     | 51,910,599/124,439,808 | 58.3%        | 62,419,226/124,439,808 | 49.8%           |

## Performance on the shared 500-example test set

| Circuit                        | Accuracy | Exact | Pairwise | KL    | Logit diff | Faithfulness |
|--------------------------------|----------|-------|----------|-------|------------|--------------|
| Normal - KL matched (ep. 80)   | 98.4%    | 98.4% | 100.0%   | 0.154 | 6.414      | 1.503        |
| Normal - epoch matched (final) | 97.2%    | 97.2% | 100.0%   | 0.194 | 6.525      | 1.529        |
| Position-aware - final         | 98.8%    | 98.8% | 100.0%   | 0.143 | 6.979      | 1.635        |

Open/total counts are hierarchy-finalized effective slots; closed columns are percentages. Position totals include nine logical sections.

# Position-Aware Node Pruning vs PEAP

Final position-aware node checkpoint; two retained PEAP operating points

## Circuit sizes

| Circuit             | Abstract edges active/full | Edges closed | High-level nodes         | Native PEAP token edges |
|---------------------|----------------------------|--------------|--------------------------|-------------------------|
| Position-aware node | 1,276/109,722              | 98.84%       | 100/1404 component slots | --                      |
| PEAP top-5k         | 4,529/109,722              | 95.87%       | 734 selected nodes       | 12,303.8/590,076.2      |
| PEAP top-10k        | 9,369/109,722              | 91.46%       | 1115 selected nodes      | 27,316.3/590,076.2      |

## Performance on the shared 500-example test set

| Circuit             | Accuracy | Exact | Pairwise | KL    | Logit diff | Faithfulness |
|---------------------|----------|-------|----------|-------|------------|--------------|
| Position-aware node | 98.8%    | 98.8% | 100.0%   | 0.143 | 6.979      | 1.635        |
| PEAP top-5k         | 89.2%    | 89.2% | 99.4%    | 0.100 | 3.637      | 0.852        |
| PEAP top-10k        | 97.4%    | 97.4% | 99.4%    | 0.029 | 3.861      | 0.905        |

Size rule: surviving nodes imply every valid incident edge, following circuit\_pruning-argo. For PEAP compatibility, Q/K/V are collapsed into one abstract connection and causal logical-position transport is included.

KL is  $D_{KL}(\text{full} \parallel \text{circuit})$ ; logit difference is IO minus distractor; faithfulness is circuit/full logit difference.